

AAA (AUTHENTICATION, AUTHORIZATION,ACCOUNTING)

- 1. Write a short scenario where a user tries to access a private WiFi network (like at a campus or airport) and describe how each of the three AAA functions—Authentication, Authorization, and Accounting—would apply step by step in this situation.**

Scenario:

A student at a college tries to connect their laptop to the college's private Wi-Fi network.

1. Authentication:

The student enters their college username and password. The Wi-Fi system sends these details to the AAA server, which verifies whether the username and password are correct.

2. Authorization:

After successful authentication, the AAA server checks what the student is allowed to access. For example, the student may be allowed to use the internet and college learning websites but may not have access to administrative systems.

3. Accounting:

The AAA server records information such as the student's username, login time, logout time, IP address, and amount of network usage. This information can be used for security monitoring and troubleshooting.

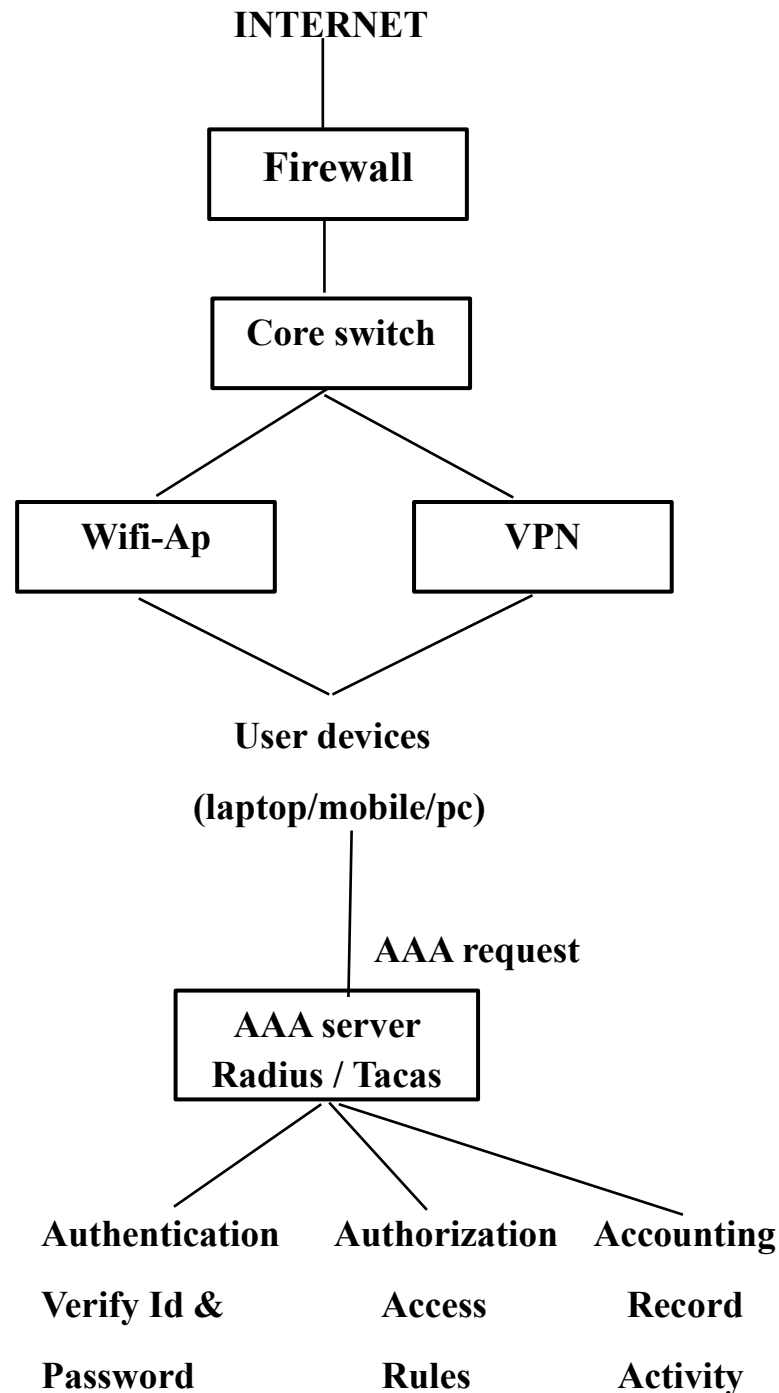
Flow:

Connect Wi-Fi → Authentication → Authorization → Network Access → Accounting/Logging

2. Create a table comparing at least two different AAA deployment models (such as centralized vs distributed), listing three advantages and three disadvantages of each for a company like Zomato that handles sensitive user data.

Feature	Centralized AAA	Distributed AAA
Meaning	All AAA requests are handled by one or a small group of central AAA servers.	AAA services are provided by multiple servers at different locations.
Advantage 1	Easy to manage from one location.	Reduces dependency on one central location.
Advantage 2	Policies can be applied consistently.	Can provide faster authentication for users in different locations.
Advantage 3	Easier monitoring and reporting.	Better availability if one server becomes unavailable.
Disadvantage 1	Central server failure can affect many users.	More difficult to manage and maintain.
Disadvantage 2	Can become a performance bottleneck with many requests.	Policies may become inconsistent between locations.
Disadvantage 3	Requires reliable connectivity to the central server.	Higher infrastructure and maintenance cost.

3. Draw a simple network diagram (can be hand-drawn and photographed) showing how AAA servers are integrated into a typical enterprise network for user access control, labeling where Authentication, Authorization, and Accounting processes happen.



- Authentication: AAA server verifies the user's username/password or other credentials.
- Authorization: AAA server determines which network resources the authenticated user can access.

- Accounting: AAA server records login attempts, session duration, IP address, and network activity.

**4. Pick any app you use daily (Instagram, Paytm, or Flipkart) and write a paragraph explaining how AAA concepts could be applied to secure user logins and actions within that app.

Hint: Think about what happens when you log in, what features you are allowed to use, and how your actions might be tracked for security or billing.**

- Flipkart can apply AAA concepts to protect user accounts and activities. **Authentication** happens when a user logs in using their registered mobile number, email, password, or verification code. After successful login, **Authorization** determines what the user is permitted to do, such as viewing products, placing orders, managing their own account, or accessing order history.
- **Accounting** can maintain records of activities such as login attempts, orders, payments, and account changes. These records can help with security monitoring, fraud detection, troubleshooting, and transaction history.

5. Use ChatGPT or any AI tool to generate a sample AAA policy for a college WiFi network that restricts access to students only, allows certain websites, and logs all access attempts. Paste the generated policy and briefly explain how each AAA function is addressed.

College Wi-Fi AAA Security Policy

Purpose:

The purpose of this policy is to provide secure Wi-Fi access only to registered college students and monitor network usage.

Authentication Policy:

- Only registered students can access the college Wi-Fi.
- Students must use their college username and password.
- Invalid login attempts will be rejected.

- Multiple failed login attempts may temporarily block the account.

Authorization Policy:

- Authenticated students can access the internet and approved educational websites.
- Access to restricted administrative systems is not permitted.
- Websites that are blocked by college policy will not be accessible.
- Different access permissions may be assigned to students, faculty, and administrators.

Accounting Policy:

- All successful and unsuccessful login attempts will be logged.
- The system will record username, IP address, login/logout time, and session duration.
- Logs will be stored securely and reviewed for suspicious activity.
- Network logs will be retained according to college security requirements.